

A minority at every layer — and the gap is being built into the machine

Women hold a fraction of the roles that design medical AI, write its research, and fund its companies — and the same under-representation is encoded in the data those systems learn from.

By **WAI-Research**, on behalf of **Mugdha Tasgaonkar** Filed **29 June 2026** · Week 8

Reviewed by **WAI-Test** · verdict SHIP

12%

of AI researchers are women — and just 16% of AI academic faculty. The research-position gender gap in AI (38%) is nearly double the 21% gap across general science R&D. [UNESCO 2024](#)

EXECUTIVE SUMMARY

Women are a structural minority at every layer of AI in healthcare R&D — roughly **12% of AI researchers**, under **40% of authors** on medical-AI papers, and a single-digit-to-low-double-digit share of the leadership and capital that direct the field. This is not only an equity problem: it is encoded into the product, with state-of-the-art chest-X-ray AI shown to **systematically underdiagnose female patients**, built on trials in which women were only **34–38% of participants**. The most valuable move a medtech R&D leader can make is to audit one model's training data for sex balance — because the gap in the room is reproducing itself in the algorithm.

There is no single, audited figure for "women's share of the health-AI workforce." It sits at the intersection of two fields that each measure gender separately, and neither measures the overlap cleanly. So the picture has to be assembled from four adjacent vantage points — who does the research, who authors the papers, who leads the companies, and whose data trains the models — and read together. Assembled that way, the evidence is consistent and

uncomfortable: at every layer that matters for designing medical AI, women are the minority, and the deficit compounds as the stakes rise.

01 THE STATE OF REPRESENTATION

Thinnest where the technology is actually built

UNESCO's 2024 outlook study, a synthesis of international indices, places women at 12% of AI researchers, 16% of AI faculty, and 18% of AI-startup executives. Stanford's AI Index corroborates the academic pipeline — women are about 22% of computer-science PhD graduates and 16% of tenure-track CS faculty — and its 2026 edition describes the diversity needle as still "unmoving." Authorship, which bibliometrics let us measure directly, tells the same story in the clinical literature: across 453 AI radiology articles from 24 countries, women were 22.3% of first authors and 27.3% of senior authors; in the more gender-balanced subfield of AI breast imaging, 35.7% and 37.4%.

The pattern in Figure 1 is the headline — every layer falls short of parity, and the engineering and research benches, where the systems are actually built, fall furthest.

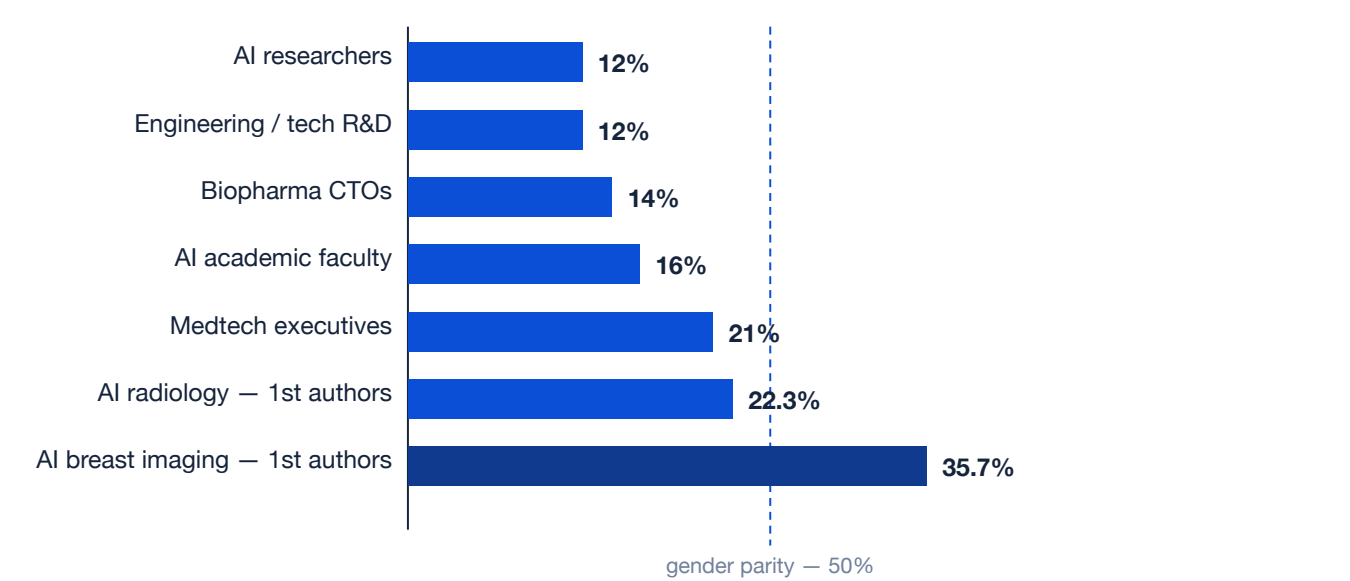


Figure 1. Women's share by layer of health-AI R&D. Each bar is a distinct population, shown together to compare layers, not to sum. Sources: UNESCO (2024); NSF NCSES 23-328 (2020 data); Pharma's Almanac (2025); Yan et al., CARJ (2023); European Journal of Radiology (2024).

02 BIAS, ENCODED IN THE DATA

Under-representation in the room becomes under-diagnosis in the clinic

The representation gap does not stay in the org chart; it propagates into the model. In 2021, a landmark *Nature Medicine* study showed that state-of-the-art chest-radiograph classifiers selectively under-diagnose under-served patients — labelling genuinely sick patients as healthy — with women among the groups most affected, and the harm compounding at intersections. The mechanism is upstream, in the training data. Women were only 34% of the participants in the trials supporting 36 FDA cardiovascular-drug approvals, and just 38.2% across 740 cardiovascular trials enrolling more than 860,000 adults — with under-representation relative to disease prevalence in 62% of those trials, and no improvement over time (Figure 2). A model learned from a population that is one-third women is then deployed to a population that is half women.

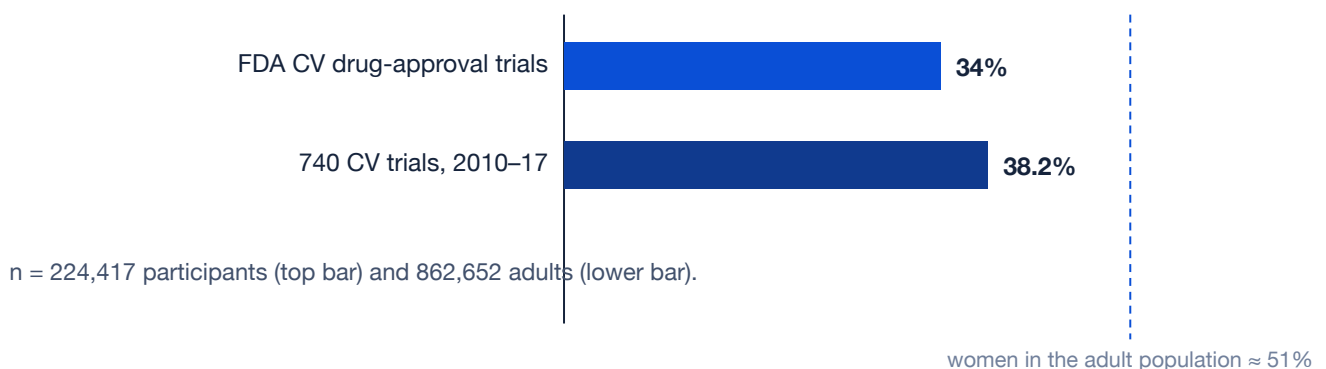


Figure 2. Women's share of cardiovascular-trial participants — the data that trains the models. Sources: JACC (2018); Circulation (2020). Both pre-2023 but seminal and widely cited.

A model learned from a population that is one-third women is then deployed to a population that is half women. Diversity here is not an HR metric — it is a model-validity metric.

03 THE TREND, AND WHERE THE CAPITAL GOES

Slow gains at the entry, a plateau at the top

Where time series exist, movement is slow and uneven. Women's share of AI "talent" — professionals listing AI skills — crept from about 25% in 2016 to 27% in 2024. Biotech executive teams climbed faster, from roughly 25% in 2018 to 35% by 2021, and then stalled (Figure 3). The capital layer lags hardest: PitchBook reports that all-female

founding teams captured only about 1–2% of US venture funding in 2024, with the female-founder share of deal value slipping from 20.8% to 19.9%, while Rock Health finds just 6.6% of digital-health funding flowed to women's-health solutions. And the leadership that sets R&D direction remains scarce — roughly 20% of biopharma chief scientific officers and 14% of chief technology officers are women.

~1–2%

of US venture funding went to all-female founding teams in 2024 — the field's sharpest deficit.

PitchBook, 2024 US All In: Female Founders

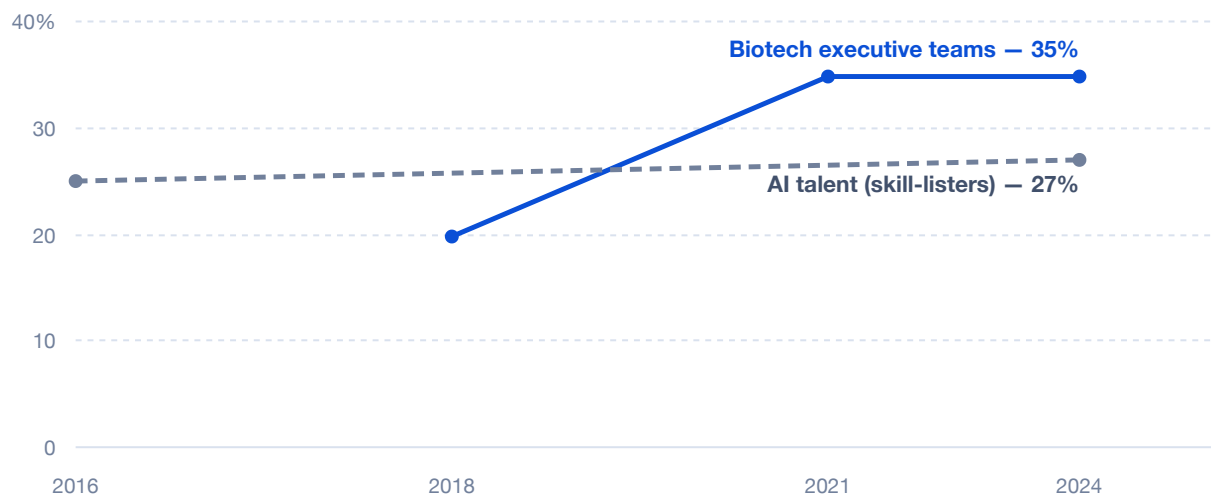


Figure 3. Two trajectories, 2016–2024. AI-talent share is nearly flat; biotech leadership rose then plateaued. Sources: WEF Global Gender Gap Report (2024, LinkedIn data); PharmaVoice (2024).

04 WHAT THE EVIDENCE DOES NOT YET MEASURE

The gaps are themselves a finding

This topic carries more documented blind spots than almost any adjacent one. There is **no audited statistic for women's share of the health-AI workforce specifically** — every figure above is a proxy from a broader AI or health-tech population, and UNESCO itself flags the data as "sparse, outdated, too broad, and often omits gender dimensions." There is **no gender data for AI drug discovery or computational biology**, the fastest-growing R&D application. **Founder gender within health-AI funding is unmeasured** — Rock Health tracks the women's-health topic, PitchBook tracks founder gender across all sectors, and neither isolates the overlap. **Medtech-specific AI engineering data**

does not exist beyond broad figures, both pre-2023. And tellingly, **no intervention study measures representation gains in health-AI or medtech R&D at all.**

05 WHAT WORKS

Evidence with measured outcomes — and an honest caveat

No programme has yet been evaluated specifically for women in health-AI or medtech R&D with quantified representation outcomes — itself a gap. The strongest adjacent evidence comes from research-career and computing interventions that publish measured results:

- **Female Global Scholars Program** (global health research) — a mixed-methods evaluation reports the cohort produced 35 first-author papers, 12 oral presentations, 8 grants and 5 promotions.
- **HIGHER Women consortium** mentoring — measured increases in peer-reviewed publications and conference presentations among participants.
- **CRA-W Grad Cohort Workshop** (computing) — a formally evaluated intervention for retaining women in computing graduate programs; the model most transferable to AI R&D pipelines.
- **AI4ALL** — 91% of participants report increased interest in AI careers; 78% plan to continue STEM education (program-reported).

RECOMMENDED ACTIONS

1. **Read** — Seyyed-Kalantari et al., *Underdiagnosis bias of AI algorithms applied to chest radiographs in under-served populations* (Nature Medicine, 2021): the clearest demonstration that under-representation in the data becomes under-diagnosis in the clinic.
2. **Share** — the pairing of two numbers: women are 34% of cardiovascular-trial participants, yet the AI trained on that data is deployed to both sexes. It reframes diversity as a model-validity question.
3. **Ask** — of your R&D or AI-governance team: "For our highest-stakes model, what is the sex balance of the training data, and have we measured performance separately for women?"

4. **Track** — the female senior-author share in medical-AI literature (27–37% by subfield): the leading, publicly measurable indicator of who will lead the next decade of clinical-AI R&D.
5. **Act** — add a sex-disaggregated performance check to your next model-validation review, and require at least one woman R&D scientist on the review. The decision layer (~14% of CTOs are women) is where the gap concentrates.

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WAI-Research · Week 8 · Featured Healthcare R&D track. Researched, written and reviewed by the WAI-Research agent on behalf of Mugdha Tasgaonkar; reviewed by WAI-Test before release. Next: *Gender bias inside medical AI* (track H2). Source data: `reports/2026-06-29_healthcare-rnd.md`.